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### Hybrid installation apparatus and processes

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#### Abstract

A hybrid conductor puller-tensioner and control system are disclosed. The puller-tensioner includes a motor/generator mechanically coupled to a drum. The motor/generator is in electrical communication with a generator and a rechargeable power source. In a tensioning mode, the motor/generator can be used to charge the rechargeable power source. In a pulling mode, when the power demanded by the motor exceeds the capacity of the generator, the motor can be connected to both the generator and the rechargeable power source simultaneously to provide additional power.

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References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name  | U.S. Cl.    | CPC        |
|--------------|-------------|----------------|-------------|------------|
| 3319935      | 12/1966     | Eitel et al.   | N/A         | N/A        |
| 5826860      | 12/1997     | Theurer        | 242/397.3   | B60M 1/28  |
| 12244131     | 12/2024     | Jackson        | N/A         | H02G 1/04  |
| 2009/0267037 | 12/2008     | Johnson et al. | N/A         | N/A        |
| 2018/0313882 | 12/2017     | McNulty et al. | N/A         | N/A        |
| 2023/0132039 | 12/2022     | MacDonald      | 254/134.3 R | H02G 1/04  |
| 2023/0187914 | 12/2022     | Muehlhans      | 254/134.3 R | H02G 1/08  |
| 2024/0063615 | 12/2023     | Post           | N/A         | H02G 1/081 |

FOREIGN PATENT DOCUMENTS

| Patent No.    | Application Date | Country | CPC       |
|---------------|------------------|---------|-----------|
| 109667814     | 12/2018          | CN      | N/A       |
| 2019148068    | 12/2018          | WO      | N/A       |
| WO-2021163390 | 12/2020          | WO      | H02G 1/04 |

OTHER PUBLICATIONS

Extended European Search Report in related EP Application No. 21754099.6 mailed Feb. 9, 2024. cited by applicant  
International Preliminary Report on Patentability in related PCT Application No. PCT/US2021/017738, mailed Aug. 11, 2022. cited by applicant  
International Search Report and Written Opinion in priority PCT Application No. PCT/US2021/017738, mailed Apr. 22, 2021. cited by applicant

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a § 371 national stage of International Application PCT/US2021/017738, with an international filing date of 11 Feb. 2021, which claims the benefit of U.S. Provisional Patent Application No. 62/972,800 filed 11 Feb. 2020, the benefit of the earlier filing date of which is hereby claimed under 35 USC § 119 (e). The entire contents and substance of all applications are hereby incorporated by reference.

### STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not Applicable

### THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

(2) Not Applicable

### SEQUENCE LISTING

(3) Not Applicable

### STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR A JOINT INVENTOR

(4) Not Applicable

### BACKGROUND OF THE DISCLOSURE

#### 1. Field of the Invention

(5) The invention is in the field of cable stringing apparatuses and processes, and, more particularly, to hybrid systems and methods.

#### 2. Description of Related Art

(6) High voltage utility transmission lines can transmit power over hundreds of miles with minimal losses because of the very high voltages used. Step-up transformers located at utility power generation plants increase the voltage transmission levels which minimizes losses due to the resistance of the transmission line (i.e., the conductor). As electrical demand continues to grow, higher-capacity lines and/or additional lines are needed.

(7) Equipment and tools for overhead and underground power line installation and maintenance include pullers, tensioners, bundle blocks, reel trailers, and battery tools.

(8) Exemplary equipment includes, among others, Sherman+Reilly's PT-3000 Puller Tensioner combining a puller, tensioner, and reconductor in single unit for overhead or underground applications. It has dual overhead and underground hydraulic levelwind, a direct drive hydrostatic motor, and a mechanical spline engagement system. It is capable of pulling 3,000 lbs. and tensioning 2,000 lbs.

(9) The PTX-3500 Puller Tensioner has a fully hydraulic direct drive system, and is capable of pulling up to 3,500 lbs. with additional re-conductoring capabilities, and tensioning up to 2,000 lbs. Hydraulic motor optimization allows for low force pull off at higher speeds, and hydraulic levelwind incorporates two cylinders that allow the levelwind to move smoothly and efficiently.

(10) The P-1400X Puller is a puller and reconductor in one unit with a pulling capacity of 14,000 lbs.

(11) The Sherman+Reilly BWHT 1485 Tensioner is completely hydraulic, equipped with either gas or diesel power. In a tensioning mode, they maintain a positive controlled pressure during the tensioning operation. The unit will also operate in either direction and may be used for either reeving, paying out or pulling back on the conductor. The system is equipped with a spring applied emergency brake which will automatically apply if hydraulic pressure is lost for any reason. Brakes may be applied manually when desired to park the system. The direction control lever is used to select either the Pay Out (reeve), Tension (neutral) or Pull Back mode. The line tension control is

used to regulate the conductor tension during stringing. The hydraulic pressure gauge is used to display the hydraulic system pressure during operation.

(12) Stringing high-voltage conductor lines across significant distances requires the use of conductor stringing apparatuses. The installation of power transmission lines, sometimes referred to as “pulling conductors”, or “tension stringing” utilizes a number of components spread over a wide area. A device called a conductor or cable puller-tensioner is used, although those of skill in the art know that other terms are used for this equipment. The equipment is typically termed by what it does.

(13) The stringing equipment typically work in pairs: a puller for pulling a cable element (conductors, fiber optic cables, and the like) through stringing sheaves of utility structures, like poles and towers, and a tensioner providing resistance to the cable. The paired equipment can be designed for only its purpose—a puller that only performs pulling and a tensioner that only provides tension, or one of both of the pair of equipment can be a unit capable of performing both as a puller and a tensioner, and during any stringing operation performing its specific set of tasks depending on which side of the stringing operation they are located.

(14) For simplicity, as used herein the term “puller-tensioner” includes units that only function as a puller, units that only function as a tensioner, and units that can function as both a puller and a tensioner. When appropriate for context, the term “puller” and/or “tensioner” is also used.

(15) The installation of power transmission lines, sometimes referred to as “pulling conductors”, or “tension stringing” utilizes a number of components spread over a wide area. The conductor is pulled into position under tension. As discussed, stringing equipment typically work in pairs: a puller-tensioner for pulling the conductor through stringing sheaves of utility structures, like poles and towers, and a puller-tensioner providing resistance to the conductor (and the pulling operation) until installation of the conductor is complete.

(16) A puller is set up at one end of the line section to perform the pulling operation, and a tensioner to perform the tensioning operation at the other end. Typically, a reel of conductor is staged behind the tensioner. A pulling line is strung from the puller, through stringing blocks between the puller and tensioner. The end of the pulling line is then attached to the conductor end after it has been threaded through the tensioner.

(17) During the stringing process, the conductor is pulled through the stringing sheaves until the end reaches the puller. The tension maintained between the tensioner and the puller keeps the conductor clear of the ground and other obstructions that could cause damage.

(18) A puller-tensioner is provided with one or more drive lines (for example, depending on the machine, drums, spools, bull wheels, screwing systems, other pulling systems, pulling system and the like), a drive line for each cable to be strung. The drive line of the puller is equipped with power for pulling with a force greater than the braking action of the tensioner at the other end of the line. The “pulling” can be via rotation of a drum, or via alternating action for piling the conductor, or a hand-over-hand pull, or other actions that pull the conductor.

(19) There are scenarios with the “pulling” action can be a “pushing action,” and the pushed element (cable/conductor) a pulled element (like optical ground wire).

(20) The length of the conductor being pulled/installed can be over a mile long. A running board can also be attached to the conductors, which can in turn be attached to the pulling line that pulls the running board and conductors through the stringing sheaves of the utility structures.

(21) Conventional pullers and tensioners are powered by internal combustion engines driving a hydraulic system that with appropriate gearing rotates the drive line at the specified torque and speed to pull the pulling line/conductor. Tensioning is performed by controlling the hydraulic pressure within the tensioner's system and/or via mechanical brakes, resisting the rotation of the tensioner's drive line (being for example a drum) and creating the desired tension in the conductor line.

(22) Reliance only on internal combustion engines for power supporting puller-tensioners present a

number of drawbacks. Internal combustion engines are noisy and thus disadvantageous when used in areas that govern limited noise pollution. Internal combustion engines are relatively heavy polluters and thus disadvantageous to good environmental stewardship. Additionally, the drive output of an internal combustion engine is tied to the combustion engine's torque curve, which can be disadvantageous when used with pullers and tensioners.

(23) It is, therefore, desirable to augment power supplied conventionally only by internal combustion engines/generators if not remove completely the dependence upon internal combustion engines in pushing and/or pulling operations. To the extent it remains beneficial to have pushing and/or pulling equipment supplied with both an engine and electric power in appropriate situations, it is an object of the present invention to combine both engine technologies with electric technologies to provide a hybrid puller-tensioner and processes regarding same. Aspects of the present disclosure address these and other issues.

#### BRIEF SUMMARY OF THE INVENTION

(24) In accordance with an exemplary embodiment of the present invention, a puller-tensioner comprises an engine, (for example, thermal engines, internal combustion engines, external combustion engines, reaction engines, fuel cells and the like), a generator to convert power from the engine into electric power, a rechargeable power source, a drive line, and a motor/generator, coupled to the drive line, and in electrical communication with the generator and the rechargeable power source, the motor/generator driving the drive line in a pulling mode, and resisting the driving of the drive line in a tensioning mode, wherein, in the pulling mode, the motor/generator is configured to receive power from the rechargeable power source and/or from the generator, and wherein, in the tensioning mode, the rechargeable power source is configured to receive power from the motor/generator.

(25) Thus, an exemplary puller-tensioner is a hybrid, having two (or more) different sources of power, for example a battery/capacitor bank and a generator/engine system. Both sources are preferably configured to operate the pushing and/or pulling operation solely, at least for a period of time, should one source of power not be available.

(26) For example, in a first situation, a puller-tensioner may be in a low noise environment and/or a clean air environment, where running an engine like an internal combustion engine is disfavored if not banned. Thus, the battery/capacitor bank may need sufficient capacity to operate the pushing and/or pulling operation solely, at least until the puller-tensioner is redeployed in an area where the internal combustion engine can be used. Or, the battery/capacitor bank should be capable of recharging, including from the motor during tensioning or from other sources, in order to maintain a capacity to operate the pushing and/or pulling operation.

(27) Of course, to be successful in this first situation, the battery/capacitor bank must be able to handle the pushing and/or pulling operation (managing the drive line) on its own, and thus the puller-tensioner must be in an area where the battery/capacitor bank can be recharged solely by on-board recharging, but without running an internal combustion engine, and/or recharged solely by an external power supply (in range of power from the grid for example), or a combination of both on-board recharging and external recharging.

(28) In a second situation, the puller-tensioner may be in a low noise environment and/or a clean air environment, where running an internal combustion engine is disfavored but not banned. Thus, the battery/capacitor bank is the preferred source to operate the pushing and/or pulling operation, and the internal combustion engine/generator a less preferred source.

(29) To be successful in this second situation, when the battery/capacitor bank is not enough/no longer enough to handle the pushing and/or pulling operation, the generator is used to power the motor to handle the pushing and/or pulling operation and/or as a resource for recharging the battery/capacitor bank. The engine can also be directly communitive with the drive line, and thus complete the pushing and/or pulling operation in this way.

(30) In a third situation, the puller-tensioner may be in an environment where running a, internal

combustion engine is useful and allowed. Thus, both the battery/capacitor bank and the internal combustion engine/generator are managed to optimize use of fuel/stored electricity with primary concerns related to beneficial pricing of running one power source and another power source in beneficial ways and/or related to beneficial power needs of a particular pushing and/or pulling operation while running one power source and another power source in beneficial ways.

(31) Under any of the above approaches, battery/capacitor banks have their own disadvantages. Batteries/capacitor banks are limited in the amount of energy they can provide. In addition, because cables are often strung in remote locations, it may be difficult to locate a suitable rechargeable source to recharge batteries/capacitor banks. As a result, an all/only electric puller-tensioner might be useful only for short-durations or where batteries are very large to provide sufficient capacity.

(32) In the pulling mode, the motor/generator can be configured to receive power from only the rechargeable power source.

(33) The puller-tensioner can further comprise a kinetic energy store to convert electrical energy from the motor/generator to mechanical energy during a tensioning operation.

(34) The puller-tensioner can further comprise an external power connection to provide power to the motor/generator.

(35) The puller-tensioner can further comprise a resistor bank to dissipate electric energy received from the motor/generator during a tensioning operation.

(36) In accordance with another exemplary embodiment of the present invention, an innovative user interface and intelligent control can be provided to both provide an operator with real-time information on the various systems on the puller-tensioner and whether any one or more power sources will be sufficient to complete the stringing operation, warn the operator if conditions warrant a warning, and control/throttle the driving and the resisting of the drive line to lengthen the amount of time current sources can continue the stringing operation and/or automatically pull power from alternative sources to be sure a pull can be completed.

(37) For example, the present invention can include a method of providing an operator with information related to the control of a drive line of a puller-tensioner during a stringing operation comprising driving the drive line in a pulling mode of the stringing operation by a motor during a pulling period, wherein the motor is powered by a first source of power, resisting the drive line in a tensioning mode of the stringing operation by the motor during a tensioning period, monitoring the driving and resisting to determine if the capacity of the battery/capacitor bank can complete the stringing operation, and providing an operator with data sufficient to manage the driving and resisting, and the provision of any additional power needs to complete the stringing operation.

(38) The data can be displayed for the operator.

(39) The method can further comprise alerting the operator if the capacity of the battery/capacitor bank cannot complete the stringing operation.

(40) In accordance with another exemplary embodiment of the present invention, the present invention can intelligently control a puller among a parallel configuration, a series configuration, and/or a series-parallel configuration with regarding to hybrid power sourcing.

(41) A method of operating a drive line of a puller during a stringing operation can comprise driving the drive line in a pulling mode of the stringing operation during a pulling period, and selectively driving the drive line in the pulling mode among at least a parallel configuration, a series configuration, and/or a series-parallel configuration, wherein in the parallel configuration, a motor located on the puller-tensioner and an engine located on the puller-tensioner can both individually drive the drive line or both can jointly drive the drive line, the motor having a first source of power comprising a battery/capacitor bank, wherein in the series configuration, the motor individually drives the drive line, the motor having the first source of power and a second source of power, the first source of power comprising the battery/capacitor bank and the second source of power comprising a generator powered by the engine, and wherein in the series-parallel configuration the motor and engine can both individually drive the drive line or both can jointly

drive the drive line, and the motor has the first source of power and the second source of power, the first source of power comprising the battery/capacitor bank and the second source of power comprising the generator powered by the engine.

(42) In accordance with another exemplary embodiment of the present invention, a method of operating a drive line of a puller during a stringing operation comprises driving the drive line in a pulling mode of the stringing operation by a motor during a pulling period, and powering the motor by a first source of power located on the puller, wherein the first source of power comprises a battery/capacitor bank.

(43) During the pulling period, the powering can be solely by the first source of power.

(44) The method can further comprise recharging the first source of power.

(45) The method can further comprise powering the motor by a second source of power located on the puller different than the first source.

(46) Recharging the first source of power can comprises recharging with a second source of power located on the puller different than the first source, wherein the second source of power comprises a generator in electrical communication with the first source of power, and wherein the generator converts power from an engine into power.

(47) Recharging the first source of power can comprises recharging with a power source not located on the puller.

(48) Recharging the first source of power can comprises recharging with a kinetic store.

(49) During a first portion of the pulling period, the powering can be solely by the first source of power, and wherein during a second portion of the pulling period different than the first portion, the powering is solely by the second source of power.

(50) During at least a portion of the pulling period, the powering can be concurrently by the first source of power and by the second source of power.

(51) The second source of power can comprise a generator in electrical communication with the first source of power, and wherein the generator converts power from an engine into power.

(52) In accordance with another exemplary embodiment of the present invention, a method of operating a drive line of a puller-tensioner during a stringing operation comprises driving the drive line in a pulling mode of the stringing operation by a motor during a pulling period, resisting the drive line in a tensioning mode of the stringing operation by the motor during a tensioning period, and powering the motor by a first source of power located on the puller, wherein the first source of power comprises a battery/capacitor bank.

(53) The puller-tensioner is selected from the group consisting of overhead pullers, overhead tensioners, bull wheel pullers, bull wheel tensioners, overhead puller-tensioners, underground pullers, v-groove pullers, v-groove puller-tensioner, and pilot line winders.

(54) During the pulling period, the powering can be solely by the first source of power.

(55) The method can further comprise recharging the first source of power.

(56) The method can further comprise powering the motor by a second source of power located on the puller different than the first source.

(57) Recharging the first source of power can comprise recharging with a second source of power located on the puller different than the first source, wherein the second source of power comprises a generator in electrical communication with the first source of power, and wherein the generator converts power from an engine into power.

(58) Recharging the first source of power can comprise recharging with a power source not located on the puller.

(59) Recharging the first source of power can comprise recharging with a kinetic store.

(60) During a first portion of the pulling period, the powering can be solely by the first source of power, and wherein during a second portion of the pulling period different than the first portion, the powering can be solely by the second source of power.

(61) During at least a portion of the pulling period, the powering can be concurrently by the first

source of power and by the second source of power.

(62) The second source of power can comprise a generator in electrical communication with the first source of power, and wherein the generator converts power from an engine into power.

(63) In accordance with another exemplary embodiment of the present invention, a method of operating a drive line of a puller-tensioner during a stringing operation comprises driving the drive line in a pulling mode of the stringing operation during a pulling period, resisting the drive line in a tensioning mode of the stringing operation during a tensioning period, and selectively driving the drive line in the pulling mode among at least a parallel configuration, a series configuration, and/or a series-parallel configuration, wherein in the parallel configuration, a motor located on the puller-tensioner and an engine located on the puller-tensioner can both individually drive the drive line or both can jointly drive the drive line, the motor having a first source of power comprising a battery/capacitor bank, wherein in the series configuration, the motor individually drives the drive line, the motor having the first source of power and a second source of power, the first source of power comprising the battery/capacitor bank and the second source of power comprising a generator powered by the engine, and wherein in the series-parallel configuration the motor and engine can both individually drive the drive line or both can jointly drive the drive line, and the motor has the first source of power and the second source of power, the first source of power comprising the battery/capacitor bank and the second source of power comprising the generator powered by the engine.

(64) In accordance with another exemplary embodiment of the present invention, a method of controlling a drive line of a puller-tensioner during a stringing operation comprises driving the drive line in a pulling mode of the stringing operation by a motor during a pulling period, wherein the motor is powered by a first source of power, resisting the drive line in a tensioning mode of the stringing operation by the motor during a tensioning period, and intelligently controlling the driving and resisting, wherein the first source of power comprises a battery/capacitor bank, wherein at least a portion of energy produced during the resisting is stored in the battery/capacitor bank, and wherein intelligently controlling comprises determining an amount of the stringing operation completed, determining an amount of the stringing operation left to be completed, monitoring a current capacity of the first power source, determining if the current capacity of the first power source and the added capacity from one or more tensioning periods in the amount of the stringing operation left to be completed is sufficient to power the motor during one or more pulling periods in the amount of the stringing operation left to be completed, changing one or more driving characteristics and resisting characteristics to extend the use of the first source of power to complete as much of the stringing operation before battery/capacitor bank exhaustion.

(65) Intelligently controlling can further comprise providing sufficient power to the drive line from a second source of power different than the first if the battery/capacitor bank will exhaust prior to completion of the stringing operation, and completing the stringing operation with the second source of power and/or augmenting the use of the battery/capacitor bank with the second source of power to complete the stringing operation.

(66) Intelligently controlling can further comprise providing sufficient additional power to the battery/capacitor bank so the battery/capacitor bank will not exhaust prior to completion of the stringing operation.

(67) The providing can be from a generator powered by an engine.

(68) The providing can from an external source of power.

(69) In accordance with another exemplary embodiment of the present invention, a puller comprises an engine, a generator to convert power from the engine into electric power, a rechargeable power source, a drive line, and a motor/generator, coupled to the drive line, and in electrical communication with the generator and the rechargeable power source, the motor/generator driving the drive line in a pulling mode.

(70) The engine can be coupled to the drive line.



- (71) In the pulling mode, the motor/generator can be configured to receive power from only the rechargeable power source.
- (72) In the pulling mode, the motor/generator can be configured to receive power from the rechargeable power source and from the generator.
- (73) The rechargeable power source can be configured to receive power from the motor/generator.
- (74) The rechargeable power source can be configured to receive power from the generator.
- (75) The rechargeable power source can be configured to receive power from the motor/generator and the generator.
- (76) In accordance with another exemplary embodiment of the present invention, a puller comprises an engine, a generator to convert power from the engine into electric power, a rechargeable power source, a drive line, and a motor/generator, coupled to the drive line, and in electrical communication with the generator and the rechargeable power source, the motor/generator driving the drive line in a pulling mode, wherein the engine is coupled to the drive line, wherein, in the pulling mode, the motor/generator is configured to receive power from the rechargeable power source and from the generator.
- (77) The rechargeable power source can be configured to receive power from the motor/generator and from the generator.
- (78) In accordance with another exemplary embodiment of the present invention, a method of operating of a puller-tensioner comprising an engine, a generator, a motor/generator, a rechargeable energy store, and a drum, comprises operating the engine to cause the generator to generate power, driving the motor/generator with the power to control the driving of the drum, determining that an power demand for a current operation of the motor/generator exceeds a maximum output of the generator, stopping the motor/generator, sending the power from the generator to the rechargeable energy store to charge the rechargeable energy store, determining that the rechargeable energy store is charged to a predetermined level, and driving the motor/generator with electric energy from both the generator and the rechargeable energy store at the same time.
- (79) These and other objects, features and advantages of the present invention will become more apparent upon reading the following specification in conjunction with the accompanying drawing figures.
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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

- (1) The accompanying figures, which are incorporated in and constitute a part of this specification, illustrate several aspects described below.
- (2) FIG. 1 depicts conductor installation with a pair of puller-tensioners.
- (3) FIG. 2 is a side view of an example puller-tensioner.
- (4) FIG. 3 is a block diagram of an example power assembly and control system.
- (5) FIG. 4 is a flowchart of an example conductor installation method.
- (6) FIG. 5 is a flowchart of an example conductor installation method.

### **DETAILED DESCRIPTION OF THE INVENTION**

(7) To facilitate an understanding of the principles and features of the various embodiments of the invention, various illustrative embodiments are explained below. Although exemplary embodiments of the invention are explained in detail, it is to be understood that other embodiments are contemplated. Accordingly, it is not intended that the invention is limited in its scope to the details of construction and arrangement of components set forth in the following description or illustrated in the drawings. The invention is capable of other embodiments and of being practiced or carried out in various ways. Also, in describing the exemplary embodiments, specific terminology will be resorted to for the sake of clarity.

(8) It must also be noted that, as used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural references unless the context clearly dictates otherwise. For example, reference to a component is intended also to include composition of a plurality of components. References to a composition containing “a” constituent is intended to include other constituents in addition to the one named.

(9) Also, in describing the exemplary embodiments, terminology will be resorted to for the sake of clarity. It is intended that each term contemplates its broadest meaning as understood by those skilled in the art and includes all technical equivalents which operate in a similar manner to accomplish a similar purpose.

(10) Ranges may be expressed herein as from “about” or “approximately” or “substantially” one particular value and/or to “about” or “approximately” or “substantially” another particular value. When such a range is expressed, other exemplary embodiments include from the one particular value and/or to the other particular value.

(11) Similarly, as used herein, “substantially free” of something, or “substantially pure”, and like characterizations, can include both being “at least substantially free” of something, or “at least substantially pure”, and being “completely free” of something, or “completely pure”.

(12) By “comprising” or “containing” or “including” is meant that at least the named compound, element, particle, or method step is present in the composition or article or method, but does not exclude the presence of other compounds, materials, particles, method steps, even if the other such compounds, material, particles, method steps have the same function as what is named.

(13) It is also to be understood that the mention of one or more method steps does not preclude the presence of additional method steps or intervening method steps between those steps expressly identified. Similarly, it is also to be understood that the mention of one or more components in a composition does not preclude the presence of additional components than those expressly identified.

(14) The materials described as making up the various elements of the invention are intended to be illustrative and not restrictive. Many suitable materials that would perform the same or a similar function as the materials described herein are intended to be embraced within the scope of the invention. Such other materials not described herein can include, but are not limited to, for example, materials that are developed after the time of the development of the invention.

(15) Examples of the present disclosure can comprise puller-tensioners for stringing wires, pulling lines, ropes, cables, and the like (collectively, “pulling lines”). The system can comprise an engine coupled to a generator, which drives a motor and can charge a battery bank, a capacitor bank, or both (“energy bank” or “battery/capacitor bank”). The energy bank, likewise, can provide power to the motor as needed.

(16) During tensioning, the motor provides resistance to the rotation of a drive line (cable drum) to provide tension to the line for proper stringing conditions. When providing resistance, the motor can return energy to be stored in the energy bank. When the power generated, or the rate of power generated, is greater than can be absorbed by the energy bank, a resistor bank can be used to dissipate excess power.

(17) In some cases, the engine can be a combustion engine that may provide additional resistance to counteract any excess pull. During pulling, on the other hand, the motor (powered by the energy bank, as needed) turns a drum that pulls the line for installation. In some cases, however, more power is needed for installation than can be provided by the generator or energy bank alone. In this case, the generator may be used to pre-charge the energy bank to enable both the generator and the energy bank to be used in unison to provide additional power to the motor than can be provided by either source alone.

(18) In other situations, the combustion engine can be directly mechanically communicative to the drive line, assisting the motor or altogether bypassing the need for the motor in specific situations.

(19) Examples of the present disclosure may include additional energy sources and stores. For

example, the system can be adapted to receive power from an external power source (e.g., by plugging into an external generator or the power grid). The external power can directly power the motor, charge the battery/capacitor bank and other systems.

(20) In an embodiment, the puller-tensioners may include a kinetic energy store (e.g., a flywheel). The kinetic energy store may be driven during tensioning if the battery and/or capacitor bank are unable to absorb all of the energy produced. When the energy to be absorbed is reduced, the puller-tensioners may then utilize the kinetic energy store to charge the energy bank.

(21) For ease of explanation, the system is discussed below with reference to stringing and supporting power and communications lines. One of skill in the art will recognize, however, that the system is not so limited. Indeed, the system could be used in any number of industries where ropes, support cables (e.g., for ski lifts), communications cables, wires, and other similar products need to be efficiently installed and supported. Thus, the description below is intended to be illustrative and not limiting.

(22) FIG. 1 illustrates an example of conductor installation **100**. Two puller-tensioners **105** and one puller-tensioner **110** are situated at opposite ends of a span for installation. In some embodiments, the system **100** includes two tensioners **105**, a running board **114**, puller **110**, power transmission towers/poles **118a**, **118b**, one or more blocks **120a**, **120b** associated with each power transmission pole, conductors **122a**, **122b** and a pulling line **124**.

(23) The puller **110** pulls a pulling line **124** connected to the conductors **122a**, **122b** through the running board **114**. Meanwhile, tensioners **105** provides tension on the conductors **122a**, **122b** to limit improper sagging of the conductors **122a**, **122b** as they are pulled across poles **118a**, **118b**.

(24) In some applications, the power transmission towers/poles **118a**, **118b** extend over a distance of multiple miles (e.g., 4-5 miles). Prior to the installation, pulling line **124** is extended and strung through the blocks (e.g., **120a**, **120b**) associated with each of the power transmission towers/poles **118a**, **118b**.

(25) The running board **114** is connected on one side to the pulling line **124** at a location near to tensioner **105**. The conductors **122a**, **122b** to be installed are connected to the other side of the running board **114**. At this point in the conductor pull operation, the running board is pulled through the blocks **120a**, **120b** associated with each of the power transmission towers/poles **118a**, **118b**, thereby installing the conductors into place along the blocks and spanning the distance between the puller **110** and the tensioners **105**.

(26) More particularly, the puller **110** includes a motor that provides a pulling force to pulling line **124**. The tensioners **105** apply an opposite force (e.g., a tensioning force) that opposes the pulling force applied to pulling line **124** to maintain tension in the conductors **122a**, **122b** while allowing the conductors to be pulled through the blocks towards the puller **110**. In some embodiments, the tensioners **105** apply tension separately to each conductor **122a**, **122b** being pulled.

(27) FIG. 2 illustrates an exemplary puller-tensioner **200**. Puller-tensioner **200** includes a drive line **210** (a drum, bull wheel, drive line or the like), a console **220**, and a power assembly **230**. Drive line **210** may be turned during conductor installation, either by unraveling a conductor **122a** wrapped around it (when used for tensioning) or to wrap a pulling line **124** and/or pulling a conductor **122a** (when used for pulling).

(28) Console **220** may include controls for the puller-tensioner. The controls may operate the motor, selectively control tensioning or pulling, and/or select from the various power sources. In some cases, the controls may instruct the generator to charge the battery or another energy store in order to provide greater power to the motor (e.g., so that the generator and battery may provide power to the motor in unison). Power assembly **230** provides power to drum **210**.

(29) FIG. 3 is a block diagram of various components of power assembly **230** and control system **380**. The power assembly **230** can comprise a first source of electric power **340** and a second source of electric power **320** different than the first source, each of the first source **340** and the second source **320** configured to provide, on its own, sufficient power to operate a motor/generator

**330** for a drive line **210** associated with a stringing operation of one or more conductors, wherein the stringing operation comprises driving the drive line **210** in a pulling mode and resisting the drive line **210** in a tensioning mode.

(30) Power assembly **230** includes an engine **310**, a generator **320**, a motor/generator **330**, and a battery and/or capacitor bank (“energy bank **340**”). Motor/generator **330** operates drum **210**, either by controlling drum **210** to turn (e.g., during a pulling operation), or by resisting the turning of drum **210** (i.e., in a tensioning operation). Motor/generator **330** may receive power from generator **320** and/or energy bank **340**. Generator **320** generates electric power from engine **310** and provides the power to energy bank **340** and/or motor/generator **330**.

(31) During a pulling operation, engine **310** may be run to generator **320**. The power from generator **320** may be provided to motor/generator **330** to rotate drum **210**. In some cases, generator **320** may provide more power than is needed by motor/generator **330**. In this case, excess power from generator **320** may be provided to energy bank **340** to charge energy bank **340**. In some cases, generator **320** may provide insufficient energy to power motor/generator **330**. In such cases, energy bank **340** may be used to provide power to motor/generator **330**. In some instances, generator **320** may first charge energy bank **340** and then, together with energy bank **340**, provide increased power to motor/generator **330**. In this manner, more power than can be provided from either source alone (i.e., generator **320** or battery bank **340**) may be used to install the conductor. Of course, based on demand, power can also be provided only from energy bank **340** or the generator **320**.

(32) In some implementations, motor/generator **330** may receive power from an external power connection **350**. For example, motor/generator may have a 240V connector and be configured to connect with an external power source (e.g., an external generator or the power grid). In this case, power may be provided to motor/generator **330** from any combination of generator **320**, energy bank **340**, and external power connection **350**.

(33) During a tensioning operation, motor/generator **330** provides resistance to the turning of drum **210** to provide tension to the conductor line. The energy generated by motor/generator **330** during tensioning can be used to charge energy bank **340**. When energy bank **340** cannot store all of the energy produced (i.e., because it is fully charged or because the rate of energy production exceeds the charging rate of energy bank **340**), generator **320** and/or engine **310** may be used to absorb some of the energy. The generator **320** can be used to turn over the engine **310**, for example, to use the compression of the engine **310** to dissipate excess energy. Additionally, excess energy may be provided to external power connection **350** to enable the grid to absorb the energy and/or to dissipate energy within the external generator. In some implementations, a resistor bank **360** may also be provided for excess energy dissipation.

(34) Excess heat produced by the resistor bank **360** may be dissipated with a heat sink and/or fan, or used to heat other components of the power assembly **230**. Excess/waste heat can also be used to heat and/or cool on-board systems, like area(s) in proximity to the operator. This heat can be used to provide air-conditioning when needed, and heating when needed, to provide operator comfort.

(35) In some implementations, power assembly **230** may further include a kinetic energy store **370**, such as a flywheel. In this configuration, the excess energy generated during tensioning may be used to charge the kinetic energy store **370**. The kinetic energy store **370** may later be used to turn the generator **320**, for example, to charge the energy bank **340**. In some cases, a connection with kinetic energy store **370** may be bi-directional to enable excess energy to “charge” the kinetic energy store **370** (e.g., spin the flywheel), for example, and then use the stored kinetic energy in the flywheel to recharge the energy bank **340**.

(36) Control system **380** may be used to selectively control one or more of any components of the present invention. The various components can be communicative with the control system **380** via a number of ways, including wirelessly and wired. The components themselves can have autonomous control, outside of a “central” control system **380**, for example, to monitor component

features unique to that component, or in order to take actions in view of local control command.

(37) For example, control system **380** may be used to selectively control the energy sources (e.g., engine **310**, generator **320**, energy bank **340**, external power connection **350**, and kinetic energy store **370**) based on power consumption and/or storage rate. For example, the control system **380** may adjust a throttle for engine **310** and/or draw energy from energy bank **340** based on the current demand of motor/generator **330**. Similarly, the control system **380** may control a flow of energy from motor/generator **330** to the various sinks (e.g., engine **310**, generator **320**, energy bank **340**, external power connection **350**, resistor bank **360**, and kinetic energy store **370**) based on the state of the various components of power assembly **230** (e.g., based on a state charge of the energy bank **340**, the temperature of the resistor bank **360**, etc.). By combining hybrid power generation and dissipation systems, examples of the present disclosure provide increased power, greater adaptability, and enable the use of smaller engines, smaller generators, and smaller energy banks to achieve the same energy inputs and outputs.

(38) Control system **380** can integrate the operation of the drive line of a puller during a stringing operation, comprising driving the drive line in a pulling mode of the stringing operation during a pulling period, and selectively driving the drive line in the pulling mode among at least a parallel configuration, a series configuration, and/or a series-parallel configuration.

(39) In the parallel configuration, a motor located on the puller-tensioner and an engine located on the puller-tensioner can both individually drive the drive line or both can jointly drive the drive line, the motor having a first source of power comprising a battery/capacitor bank

(40) In the series configuration, the motor individually drives the drive line, the motor having the first source of power and a second source of power, the first source of power comprising the battery/capacitor bank and the second source of power comprising a generator powered by the engine.

(41) In the series-parallel configuration, the motor and engine can both individually drive the drive line or both can jointly drive the drive line, and the motor has the first source of power and the second source of power, the first source of power comprising the battery/capacitor bank and the second source of power comprising the generator powered by the engine.

(42) Control system **380** can integrate the operation of the drive line of a puller-tensioner during a stringing operation comprising driving the drive line in a pulling mode of the stringing operation during a pulling period, resisting the drive line in a tensioning mode of the stringing operation during a tensioning period, and selectively driving the drive line in the pulling mode among at least a parallel configuration, a series configuration, and/or a series-parallel configuration.

(43) In the parallel configuration, a motor located on the puller-tensioner and an engine located on the puller-tensioner can both individually drive the drive line or both can jointly drive the drive line, the motor having a first source of power comprising a battery/capacitor bank.

(44) In the series configuration, the motor individually drives the drive line, the motor having the first source of power and a second source of power, the first source of power comprising the battery/capacitor bank and the second source of power comprising a generator powered by the engine.

(45) In the series-parallel configuration, the motor and engine can both individually drive the drive line or both can jointly drive the drive line, and the motor has the first source of power and the second source of power, the first source of power comprising the battery/capacitor bank and the second source of power comprising the generator powered by the engine.

(46) Control system **380** can integrate the operation of the drive line of a puller-tensioner during a stringing operation comprising driving the drive line in a pulling mode of the stringing operation by a motor during a pulling period, wherein the motor is powered by a first source of power, resisting the drive line in a tensioning mode of the stringing operation by the motor during a tensioning period, and intelligently controlling the driving and resisting.

(47) The first source of power comprises a battery/capacitor bank, and at least a portion of energy

produced during the resisting is stored in the battery/capacitor bank.

(48) Intelligently controlling can comprises determining an amount of the stringing operation completed, determining an amount of the stringing operation left to be completed, monitoring a current capacity of the first power source, determining if the current capacity of the first power source and the added capacity from one or more tensioning periods in the amount of the stringing operation left to be completed is sufficient to power the motor during one or more pulling periods in the amount of the stringing operation left to be completed, changing one or more driving characteristics and resisting characteristics to extend the use of the first source of power to complete as much of the stringing operation before battery/capacitor bank exhaustion.

(49) Intelligently controlling can further comprise providing sufficient power to the drive line from a second source of power different than the first if the battery/capacitor bank will exhaust prior to completion of the stringing operation, and completing the stringing operation with the second source of power and/or augmenting the use of the battery/capacitor bank with the second source of power to complete the stringing operation.

(50) Intelligently controlling can further comprise providing sufficient additional power to the battery/capacitor bank so the battery/capacitor bank will not exhaust prior to completion of the stringing operation.

(51) The present invention can further provide an operator with information related to the control a drive line of a puller-tensioner during a stringing operation. The driving and resisting of the drive line are monitored to determine if the capacity of the battery/capacitor bank can complete the stringing operation.

(52) The operator is then provided with data sufficient to manage the driving and resisting, and the provision of any additional power needs to complete the stringing operation.

(53) The data can be displayed for the operator.

(54) The present invention can further comprise alerting the operator if the capacity of the battery/capacitor bank cannot complete the stringing operation.

(55) The present invention can incorporate the use of natural language control and/or alerts as the equipment is much less noisy without the engine running. Various components of the present invention can be partially or completely controlled and/or monitored by voice, and alerts can be provided by a speaking voice or other audible means.

(56) FIG. 4 is a flowchart of an example method **400** of operation of the hybrid puller and/or hybrid tensioner, each **200**. The method **400** may be controlled by the control system **380** of the hybrid puller and/or hybrid tensioner, each **200**. At **410**, the engine **310** is started. At **420**, the operation of the engine **310** causes the generator **320** to generate power. At **430**, the power from the generator **320** is used to drive the motor/generator **330** (i.e., to rotate the drum **210**).

(57) At **440**, the control system **380** can determine that the power from the generator **320** is no longer sufficient to meet the needs of the motor/generator **330**. As the cable is leaving and running through structures, the weight of the line means more tension. As the length of conductor **118** being pulled increases, for example, the tension required to maintain proper sag increases incrementally by the increased weight on installation sheaves may increase, potentially exceeding the output capacity of the generator **320**. Accordingly, at **450**, the motor/generator **330** can be stopped (i.e., the power to the motor/generator **330** can be interrupted) and the pulling/tensioning operation can be stopped. At **460**, power from the generator **320** can be diverted from the motor/generator **330** to charge the energy bank **340**).

(58) At **470**, the control system **380** can determine that the energy bank **340** is charged to a predetermined level. This may be based on a set charge level (e.g., 70%, 80%, 100%, etc.), the excess capacity required, or the calculated time required to finish the job, among other things. At **480**, when sufficiently charged, power may be provided to the motor/generator **330** from both the generator **320** and the energy bank **340**. Accordingly, the combined power available to the motor/generator **330** is greater than can be provided from either the generator **320** or the energy

bank **340** alone.

(59) FIG. 5 is a flowchart of an example method **500** of operation of the hybrid puller and/or hybrid tensioner, each **200**. The method **500** may be controlled by the control system **380** of the hybrid puller and/or hybrid tensioner, each **200**. At **510**, the engine **310** is started. At **520**, the operation of the engine **310** causes the generator **320** to generate power. At **530**, the power from the generator **320** is used to drive the motor/generator **330** (i.e., to rotate the drum **210**) and charge the energy bank **340** (e.g., charge a rechargeable energy store).

(60) At **540**, the control system **380** can determine that the power from the generator **320** is no longer sufficient to meet the needs of the motor/generator **330**. As the length of conductor **122a/122b** being pulled increases, for example, the friction caused by the increased weight on installation sheaves may increase, potentially exceeding the output capacity of the generator **320**. Accordingly, at **550**, the motor/generator **330** can be driven by power from both the generator **320** and power stored in the energy bank **340**. Accordingly, the combined power available to the motor/generator **330** is greater than can be provided from either the generator **320** or the energy bank **340** alone.

(61) At **560**, the battery/capacitor bank may be stopped (e.g., after pulling the cabling elements/conductors **122a/122b** is complete) and power from the generator **320** continue to be used to charge the energy bank **340**. At **570**, the control system **380** can determine that the energy bank **340** is charged to a predetermined level. This may be based on a set charge level (e.g., 70%, 80%, 100%, etc.) and/or a capacity of the energy bank **340**. At **580**, when sufficiently charged, the engine **310** may be stopped.

(62) The present invention can further comprise telemetry elements, and can provide fine driving/tension control, virtually on-demand/instantly, with electric control over conventional hydraulic systems.

(63) While several possible examples are disclosed above, examples of the present disclosure are not so limited. For instance, while the system is discussed above with reference to suspending power or communications lines, the system could also be used in many other industries such as transportation (e.g., towing, cables cars, street cars, and trains); rope for climbing, rigging, and boundaries; and virtually any other time lines need to be strung and/or suspended. In addition, while various features are disclosed, other designs could be used.

(64) Numerous characteristics and advantages have been set forth in the foregoing description, together with details of structure and function. While the invention has been disclosed in several forms, it will be apparent to those skilled in the art that many modifications, additions, and deletions, especially in matters of shape, size, and arrangement of parts, can be made therein without departing from the spirit and scope of the invention and its equivalents as set forth in the following claims. Therefore, other modifications or embodiments as may be suggested by the teachings herein are particularly reserved as they fall within the breadth and scope of the claims here appended.

## Claims

1. A stringing apparatus comprising: an engine; a generator to convert power from the engine into electric power; a rechargeable power source; a drive line; and a motor/generator, coupled to the drive line, and in electrical communication with the generator and the rechargeable power source, the motor/generator driving the drive line in a pulling mode.
2. The stringing apparatus of claim 1, wherein the engine is coupled to the drive line; wherein the rechargeable power source is in electrical communication with the generator; wherein the drive line is configured to receive a cable; and wherein the motor/generator is configured to drive the drive line in the pulling mode to pull the cable.
3. The stringing apparatus of claim 2, wherein, in the pulling mode, the motor/generator is

- configured to receive electric power from: only the rechargeable power source; or the rechargeable power source and from the generator.
4. The stringing apparatus of claim 2, wherein the rechargeable power source is configured to receive electric power from only the motor/generator, from only the generator, or from both the motor/generator and the generator.
  5. The stringing apparatus of claim 2, wherein the motor/generator is further configured to resist the driving of the drive line in a tensioning mode; and wherein, in the tensioning mode, the rechargeable power source is configured to receive electric power from the motor/generator.
  6. The stringing apparatus of claim 5, wherein, in the pulling mode, the motor/generator is configured to receive electric power from only the rechargeable power source.
  7. The puller tensioner stringing apparatus of claim 5 further comprising a kinetic energy store to convert electrical energy from the motor/generator to mechanical energy during a tensioning operation.
  8. The stringing apparatus of claim 5 further comprising an external power connection to provide electric power to the motor/generator.
  9. The stringing apparatus of claim 5 further comprising a resistor bank to dissipate electric energy received from the motor/generator during a tensioning operation.
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